

Ambient Clinical Intelligence & Patient Experience Analytics

Problem Statement: The Survey Bottleneck

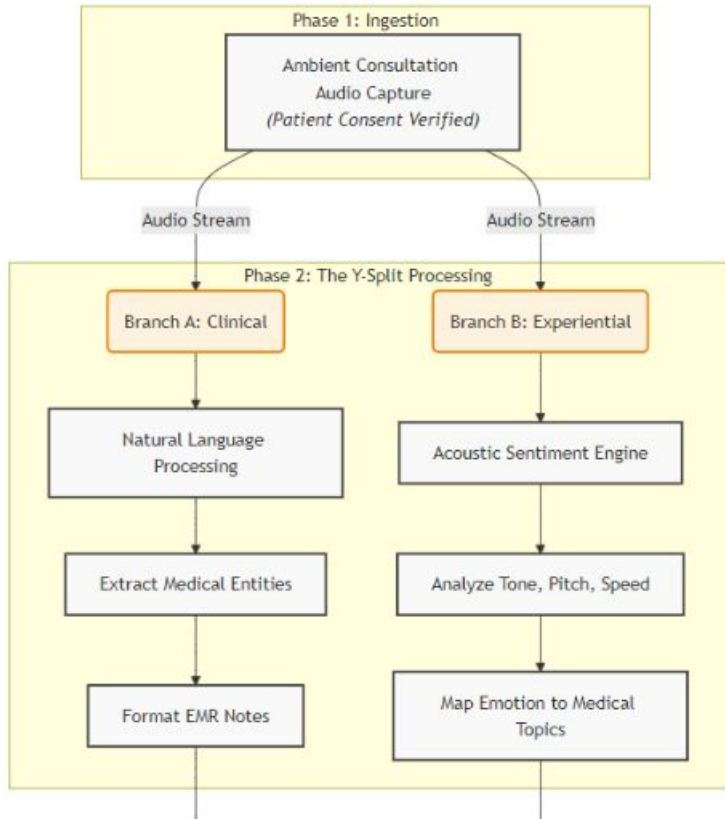
- **The Data Gap:** Current post-visit surveys yield ~5% response rates, typically capturing only extreme sentiments long after the appointment.
- **The Operational Blindspot:** Without real-time, objective data on provider interactions, leadership cannot proactively improve patient trust or treatment compliance.
- **The Goal:** Move from delayed, anecdotal complaints to a 100% passive feedback capture system.

Historical Survey Response Rate

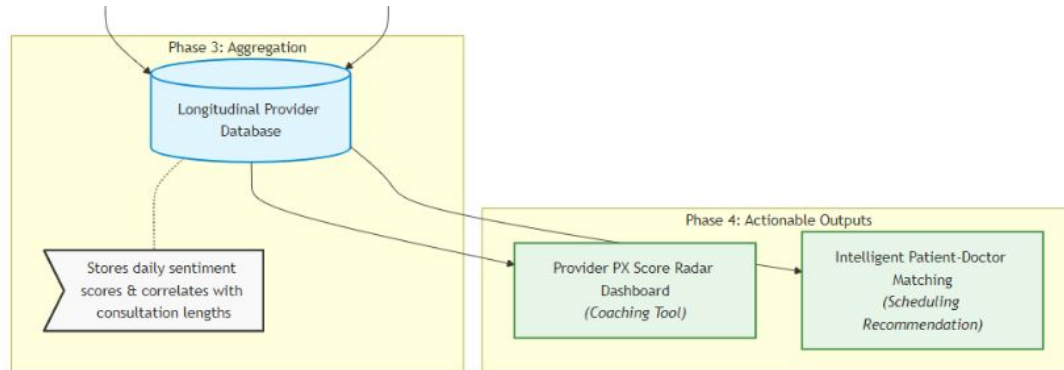
Traditional survey baseline — avg. ~5% capture (vs. 100% ambient)



Architecture



- **Data Repurposing:** Utilize the ambient consultation audio already being captured for automated EMR charting.
- **Simultaneous Processing:** A "Y-Split" data flow routes audio to both Clinical (NLP for EMRs) and Experiential processing engines simultaneously.
- **Acoustic Sentiment:** Analyzes biometric voice markers (tone, pitch, speed) to gauge patient comfort without requiring active survey participation.



Executive Impact

- **Unbiased Data:** Eliminates survey fatigue by gathering passive feedback natively during every visit.
- **Real-Time ROI:** Immediately tracks global sentiment shifts and flags critical friction alerts.
- **Longitudinal Tracking:** Replaces static data snapshots with continuous tracking of hospital-wide patient trust and comfort trends.

Executive Dashboard

CONSULTATIONS ANALYZED

12,450
↑ +12% MoM

FEEDBACK CAPTURE

100%
Natively captured

GLOBAL SENTIMENT

8.4/10
↑ 0.3 pts from last month

CRITICAL FRICTION ALERTS

14
Requires immediate review

30-Day Patient Trust Trend
Composite score derived from passive acoustic analysis - Steady upward trajectory



Acoustic Sentiment
Distribution across all 12,450 consultations



Historical Survey Response Rate
Traditional survey baseline — avg. ~5% capture (vs. 100% ambient)



Speciality Benchmarking

- **Departmental Comparison:** Aggregates performance to compare communication efficacy across different specialties (e.g., Cardiology vs. Pediatrics).
- **Targeted Interventions:** Maps specific friction points (e.g., "Wait Time Anxiety" or "Treatment Confusion") to the departments where they occur most frequently.
- **Resource Optimization:** Allows leadership to allocate coaching and scheduling resources based on objective, departmental needs.

Speciality Benchmarking Dashboard

TOP PERFORMING DEPT

Cardiology
Score: 8.7 / 10

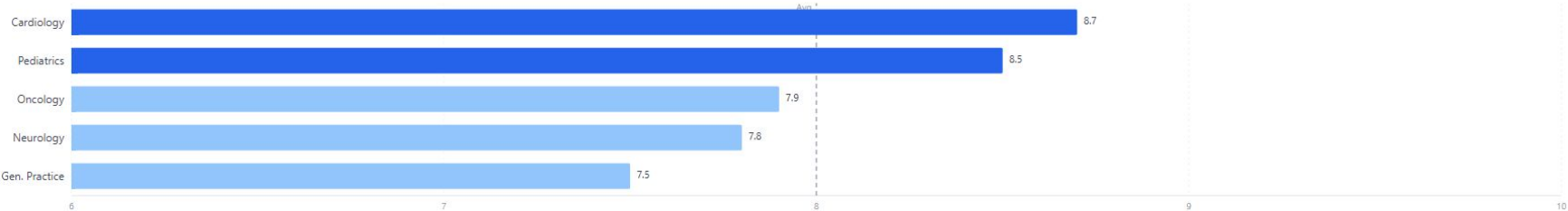
HIGHEST IMPROVEMENT

Pediatrics
↑ +1.2 pts over 60 days

HOSPITAL AVERAGE

8.0 / 10
All departments combined

Departmental PX Averages
Composite patient experience score • Dashed line = Hospital Average (8.0) • Blue = above average



Identified Friction Points

ISSUE	DEPARTMENT	SEVERITY	OCCURRENCES
Wait Time Anxiety	General Practice	High	142
Treatment Confusion	Oncology	Medium	87
Medication Instructions	Neurology	Medium	64

Provider Analytics

- **Contextual Scoring:** The PX Score Radar visually scores providers on Empathy, Clarity, and Efficiency based on acoustic data.
- **Fatigue Tracking:** Identifies interaction drop-offs (e.g., declining clarity during afternoon shifts) to adjust scheduling.

Deployment Risks & Mitigation:

- **Adoption:** Must be positioned for private professional development, not punitive monitoring.
- **Consent:** Requires seamless, non-alarming front-desk workflows.
- **Bias:** Algorithm requires human oversight to prevent socio-economic segregation in patient-matching.

Provider Dashboard

SJ

Dr. Sarah Jenkins

Cardiology · Attending Physician · Joined 2019

PATIENTS SEEN

284

AVG. CONSULT

18 min

FOLLOW-UP RATE

94%

PEER RANK

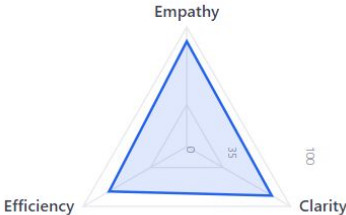
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INDIVIDUAL PX SCORE

8.7 /10

Communication Profile

Derived from patient feedback analysis — translucent blue fill, solid stroke



Empathy

88/100

Clarity

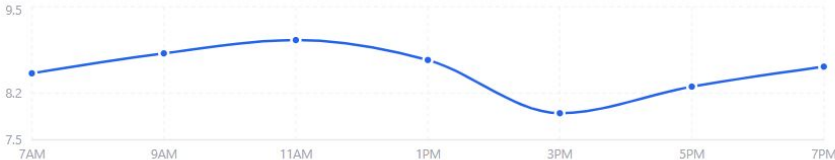
82/100

Efficiency

75/100

Sentiment Trend — Shift Timeline

PX score 7AM–7PM · Afternoon dip at 3PM flagged for coaching



Key Drivers

STRENGTHS

- Excellent Medication Explanation
- Patient Empathy & Active Listening
- Clear Follow-up Instructions

NEEDS ATTENTION

- Rushed Discharge Process
- Limited Wait Time Communication

Thank You